FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
CONCEPT RECAPITULATION TEST – I
PAPER –2
TEST DATE: 30-06-2022

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 57 questions.
- Each subject (PCM) has 19 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections**: **Section-A, Section-B & Section-C**.

  Section – A ***(01 –06, 20 – 25, 39 – 44)*:** This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

  Section – A ***(07 –10, 26 – 29, 45 – 48)*:** This section contains **SIX (06) paragraphs.** Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

  Section – B ***(11 – 13, 30 – 32, 49 – 51)*:** This section contains **NINE (09)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.**

  Section – C ***(14 – 19, 33 – 38, 52 – 57)*:** This section contains **NINE (09)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+2	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

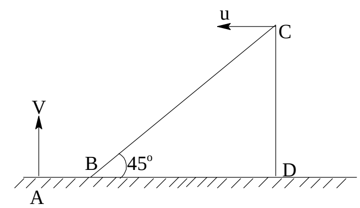
Physics

PART – I

Section – A (Maximum Marks: 24)

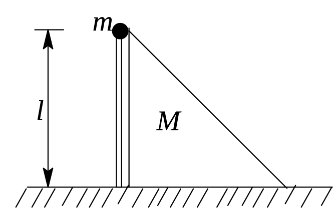
*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

1. A large wedge BCD, having its inclined surface at an angle $\theta = 45^o$ to the horizontal, is travelling horizontally leftwards with uniform velocity u=10m/s. At some instant a particle is projected vertically up with speed V=20m/s from point A on ground lying at some distance right to the lower edge b of the wedge. The particle strikes the incline BC normally, while it was falling. $[\text{g=10m/s}^2]$



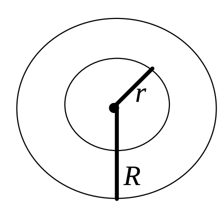
(A) The distance AB at the instant the particle was projected from A is 15m
(B) The distance of lower edge B of the wedge from point A at the instant particle strikes the incline is 15m
(C) The distance of lower edge B of the wedge from point A at the instant particle strikes the incline is 30m
(D) The path of the particle in the reference frame attached to the wedge is parabolic

2. A weightless rod of length l with a small load of mass m at one of its end is held vertical with its lower end hinged on a horizontal surface. The load touches a wedge right sets the system in motion (see figure), with rod rotating freely in vertical plane about its lower end. There is no friction. If the rod forms an angle $\theta = \pi / 3$ with the vertical at the moment the load separates from the wedge, then



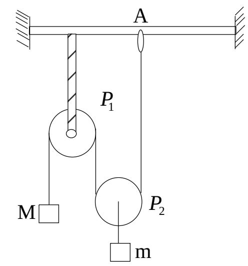
(A) The mass ratio $\frac{M}{m}$ =3
(B) The speed of the wedge at that moment will be $\sqrt{\frac{gl}{8}}$
(C) The mass ratio $\frac{M}{m}$ =4
(D) The speed of the wedge at that moment will be $\sqrt{\frac{gl}{2}}$

3. A soap bubble of radius r is formed inside another soap bubble of radius R(>r). The atmosphere pressure is P_0 and surface tension of the soap solution is T. Now the bigger bubble bursts. Assume that the excess pressure inside a bubble is small compared to P_0. For the smaller bubble choose the correct options:



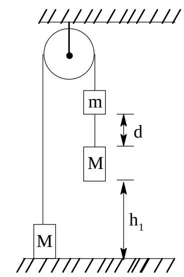
(A) The bubble shrinks
(B) The bubble expands
(C) Change in its radius is $\frac{4Tr}{3P_0R}$
(D) Change in its radius is $\frac{3Tr}{4P_0R}$

4. The arrangement shown in figure is in equilibrium with all strings vertical. The end A of the string is tied to a ring which can be slid slowly on the horizontal rod. Pulley P_1 is rigidly fixed but P_2 can move freely. A mass m is attached to the centre of pulley P_2 through a thread. Pulleys and strings are mass less.



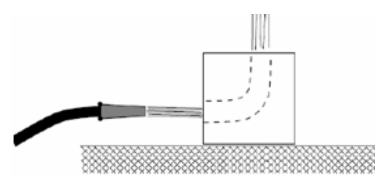
(A) Block of mass M will move up as A is moved slowly to the right
(B) Block of mass m will move up as A is moved slowly to the right
(C) It is possible, for a particle position of A, that M has no acceleration but m does have acceleration
(D) The block of mass m will have a horizontal displacement

5. There are two weights of the same mass of 2.0 kg each attached to a string looped over a braked pulley. The left weight is lying on the floor, the right one is suspended 1.0 m above the floor. There is another weight of mass 0.5 kg attached to the string 20 cm above the right weight. The string except the part around the pulley is stretched vertically. We eventually release the pulley.



(A) The speed at which the lower right weight hits the floor is 1.48 m/s
(B) The height that the left weight rises to after the lower right weight falls inelastically to the floor is 1.185 m
(C) The maximum height that the lower right weight rises back to is 1.185 m
(D) The maximum height that the lower right weight rises back to is 0.31 m

6. A block of mass M, at rest on a frictionless surface, contains a curved channel. The channel is designed so that when a horizontal jet of fluid is aimed at it, the fluid is redirected 90 and shoots straight up and out of the top of the block. The flow of the jet is Q kg/s and has a velocity v m/s. Assuming no gravity and ignoring any viscous effects. [Density of fluid is ρ]



(A) Speed of the block as a function of time is $v_b = v\left(1-e^{-\frac{\rho Qt}{M}}\right)$

(B) Speed of the block as a function of time is $v_b = v\left(\frac{\rho Qt}{\rho Qt + M}\right)$

(C) Power delivered by the fluid to the block at $t = 0$ is zero
(D) Power delivered by the fluid to the block at $t = 0$ is ρQv

Section – A (Maximum Marks: 12)

*This section contains **TWO (02) paragraphs.** Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

Paragraph for Question Nos. 07 and 08

In a homogeneous, non-magnetic, highly insulating and viscous medium, a moving particle experiences a viscous drag given by the law $\vec{f} = -b\vec{v}$. Here b is a positive constant. A particle having charge q projected with an unknown velocity from a point in the medium almost stops (its speed becomes practically negligible) after travelling a distance of 10 m in a straight line. Now a uniform magnetic field is established in the region and the same particle is again launched with the same projection velocity directed perpendicular to the magnetic field.

7. In the presence of above magnetic field, if the particle almost stops at a point 6m away from the point projection the magnetic field is equal to:

(A) $\frac{4b}{3q}$

(B) $\frac{3b}{4q}$

(C) $\frac{4q}{3b}$

(D) $\frac{3q}{4b}$

8. The strength of the magnetic field is now doubled in magnitude. How far (in meters) from the point of projection will the particle come to an almost rest?

(A) $\frac{\sqrt{73}}{2}$

(B) $\frac{20}{\sqrt{73}}$

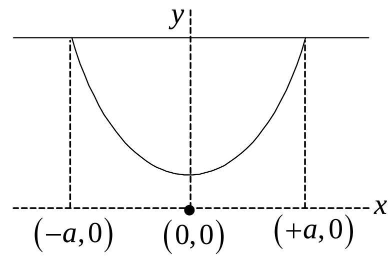
(C) $\frac{40}{\sqrt{73}}$

(D) $\frac{30}{\sqrt{73}}$

Paragraph for Question Nos. 09 and 10

When a massive string (mass/length= μ) is hung from two supports, the shape it forms is called catenary.

The equation for catenary shown in diagram is $Y = \frac{a}{2}\left[e^{-x/a} + e^{x/a}\right]$



9. Then find out tension at lowest point:
   (A) $2\mu ga$
   (B) μga
   (C) $\frac{\mu ga}{2}$
   (D) $4\mu ga$

10. The tension at the topmost points will be:
   (A) 0.77 μag
   (B) μag
   (C) $2\mu ga$
   (D) $\frac{\mu ga}{2}$

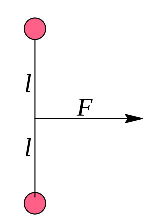
Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.***

11. Binding energy per nucleon of a fixed nucleus X^A is 6Me V. It absorbs a neutron moving with $KE = 2MeV$, and converts into Y, emitting a photon of energy $1\ MeV$. If the Binding Energy per nucleon of Y (in MeV) is found to be $\frac{(xA-1)}{(A+1)}$. Find the value of x.

12. A neutral atom of atomic mass number 100 which is stationary at the origin in gravity free emits an α-particle (A) in z-direction. The product ion is P. A uniform magnetic field exists in the x-direction. Disregard the electromagnetic interaction between A and P. If the angle of rotation of A after which A and P will meet for the first time is $6n\pi/25$ radius, what is the value of n?

13. Two identical masses, placed on a smooth horizontal surface, are connected by a light string of length 2l. A constant force F starts acting on the system in the direction shown. After some time the masses collide inelastically. The energy lost in this collision is Fl / N. Find the value of N.



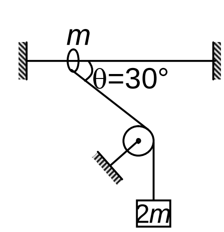
Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 14 and 15

Question Stem

A smooth ring of mass *m* can slide on a fixed horizontal rod. A massless string tied to the ring passes over a fixed smooth pulley of mass *m* and carries a block of mass 2*m* as shown in figure. At an instant the string between ring and pulley makes an angle θ = 30° with the horizontal.



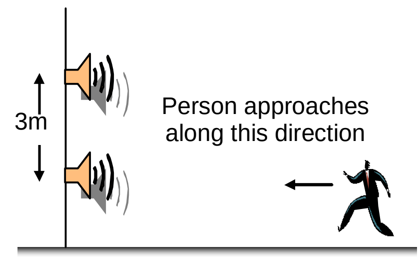
14. Acceleration of block is Kg then value of K will be:

15. Normal force due to ring on the rod is Kmg then value of K will be:

Question Stem for Question Nos. 16 and 17

Question Stem

An oscillator of frequency 680Hz drives two speakers. The speakers are fixed on a vertical pole at a distance 3m from each other as shown in the figure. A person whose height is almost the same as that of the lower speaker walks toward the lower speaker in a direction perpendicular to the pole. Assuming that there is no reflection of sound from the ground and speed of sound is $v = 340$ m/s, answer the following questions.



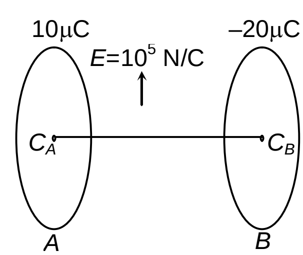
16. As the person walks toward the pole, his distance from the pole when he first hears a minimum in sound intensity is nearly (in m)

17. As the person walks toward the pole, the total number of times that the person hears a minimum in sound intensity will be.

Question Stem for Question Nos. 18 and 19

Question Stem

Two circular rings *A* and *B* each of radius *a* = 30cm are placed coaxially with their axis horizontal in a uniform electric field $E = 10^5$ N/C directed vertically upward as shown in figure. Distance between centres of the rings *A* and *B* (C_A and C_B) is 40 cm. Ring *A* has positive charge $q_A = 10\,\mu\mathrm{C}$ and *B* has a negative charge $q_B = -20\mu$C. A particle of mass *m* and charge *q* = 10μC is released from rest at the centre of ring *A*. If particle moves along C_AC_B, then (g = 10 m/s^2)



18. Mass of charge particle is (in gm)

19. Work done by electric field, when particle moves from C_A to C_B is (in Joule)

Chemistry PART – II

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

20. The ore(s) that can be concentrated by magnetic separation method are/is:
    (A) Chromite ore
    (B) Copper pyrites
    (C) Ilmenite
    (D) Cinnabar

21. Among the following the process that uses the formation of a complex is:
    (A) vapour phase refining of Nickel
    (B) separation of lead and zinc during concentration of their sulphide ores
    (C) estimation of Cu^{2+} by gravimetric analysis
    (D) oxidation of oxalic acid by $KMnO_4$

22. The complexes $[CoCl(NH_3)_5]Cl_2$ and $[Co(NH_3)_6]Cl_5$ can be distinguished by:
    (A) Magnetic moment
    (B) Depression in freezing point measurements
    (C) $AgNO_3$ test
    (D) Reaction with MnO_2 in acidic medium

23. In an octahedral complex $[\mathrm{Ma}_6]$ where all the ligands are weak field, the magnetic moment is 5.92 BM. If all the ligands are replaced by strong field ligands, the magnetic moment cannot be:
    (A) 1.73
    (B) 2.83
    (C) 5.92
    (D) 0

24. Among the following, the complexes that will not show any change in magnetic moment, on replacing a strong field ligand by weak field ligand or vice versa is/are:
    (A) $\left[\mathrm{Cr}\left(\mathrm{C_2O_4}\right)_3\right]^{3-}$
    (B) $[NiCl_4]^{2-}$
    (C) $[TiCl_6]^{2-}$
    (D) $[MnF_6]^{4-}$

25. $\mathrm{Na_2O_2}$ is used to absorption of $\mathrm{CO_2}$. If $0.78\,g$ of $\mathrm{Na_2O_2}$ is taken in a solution containing $4.4\,g$ of $\mathrm{CO_2}$, which of given statement(s) is/are correct:
    (A) $\mathrm{Na_2O_2}$ is limiting regent
    (B) $\mathrm{CO_2}$ is completely used up
    (C) 0.01 mole of salt is produced
    (D) $0.01\times N_A$ oxygen atoms will be released in gas ($\mathrm{N_A}$ is Avogadro's number)

Section – A (Maximum Marks: 12)

*This section contains **TWO (02) paragraphs.** Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

Paragraph for Question Nos. 26 and 27

A white crystalline solid A on boiling with caustic soda solution gives a gas B which on passing through an alkaline solution of potassium tetraiodomercurate (II) solution gives a brown ppt. The substance A on heating evolves a neutral gas C which is called laughing gas

26. The gas B is
(A) H_2S
(B) NH_3
(C) HCl
(D) CO_2

27. The substance A is
(A) NH_4Cl
(B) NH_4NO_3
(C) $NH_4\ NO_2$
(D) $NaNO_3$

Paragraph for Question Nos. 28 and 29

$A_{(g)} + B_{(g)} \xrightarrow{water}$ yellow turbid solution.

A turns lime water milky and both A and B are sulphur compounds which turn acidified dichromate paper green.

28. The compound A is:
(A) water insoluble
(B) used as a non aqueous solvent
(C) a gas and the similar compound on going down the group are also gases
(D) easily oxidized to a dibasic organic acid

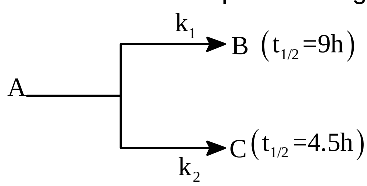
29. The hybridization of the central atom in A and B are:
(A) sp^2, sp^3
(B) sp^3, dsp^2
(C) both are sp^3 hybridized
(D) both are dsp^2 hybridized.

Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.***

30. During the discharge of a lead storage battery, the density of H_2SO_4 falls from 2.016 to $1.12\,g\,mL^{-1}$. Initially it was 25% H_2SO_4 by weight and that and finally it is $20\%\,H_2SO_4$ by weight. The battery holds $3.5L$ of acid and volume remains practically constant during the discharge. Calculate the number of ampere hours for which the battery must have been used. Given answer in nearest integer. [F=96500]

31. A radioactive isotope A undergoes simultaneous decay to yield B and C nuclei as shown:



Assuming that initially neither B nor C was present, after now many hours will the amount of C be just double to the amount of A remaining?

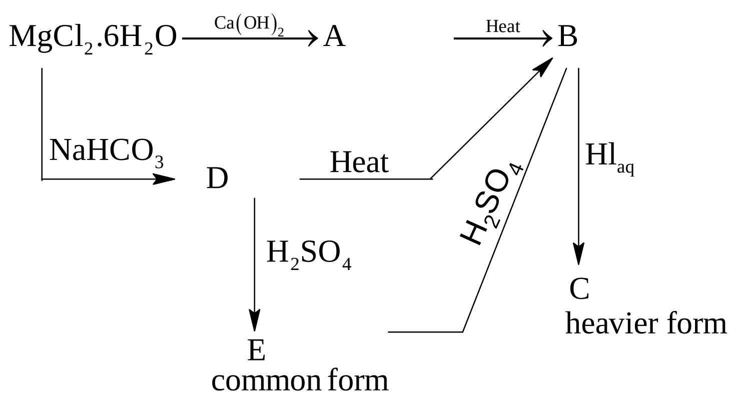
32. In the chemical reaction between stoichiometric quantities of $KMnO_4$ and KI in weakly basic solution, what is the number of moles of neutral diatomic molecule which act as an electrophile that would be released for 4 moles of $KMnO_4$ consumed?

Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 33 and 34

Question Stem



33. What is the molar mass of C by rounding molar masses of elements to nearest integer?

34. What percentage of Mg is present in compound E?

Question Stem for Question Nos. 35 and 36

Question Stem

A solution contains a mixture of $Ag^{+}\left(0.10\,M\right)$ and $Hg_2^{2+}\left(0.10\,M\right)$ which are to be separated by selective precipitation:

Hg_2I_2 $K_{sp}=2.5\times10^{-26}$ K_{sp} of $AgI=8.5\times10^{-17}$

35. Calculate the maximum concentration of iodide ion at which one of them gets precipitated almost completely. If it is A x 10^{-14} then what is A

36. What % of that metal ion that is precipitated?

Question Stem for Question Nos. 37 and 38

Question Stem

We have taken a saturated solution of $AgBr.K_{sp}$ of $AgBr$ is 12×10^{-14}. If 10^{-7} mol of $AgNO_3$ are added to $1L$ of this solution, (specific conductance) of this solution in terms of $10^{-7}\,S\,m^{-1}$ units.
Given $\lambda^0_{(Ag^+)} = 6\times10^{-3}\,S\,m^2\,mol^{-1}, \lambda^0_{(Br^+)} = 8\times10^{-3}\,S\,m^2\,mol^{-1}, \lambda^0_{(NO_3^-)} = 7\times10^{-3}\,S\,m^2\,mol^{-1}$

37. Solubility of $AgBr$ (nearest integer $a\times10^{-x}$) then what is $a+x$.

38. Find conductivity of solution (nearest integer $a\times10^{-x}$) then what is $a+x$.

Mathematics PART – III

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

39. The solution of differential equation $3\frac{dx}{dy}=\frac{x}{x^3-y}$ is $x^{\ell}=mx^n y+c$, then which of the following is/are **CORRECT**? {c is any arbitrary constant}

(A) $\ell+m+n=11$

(B) $\ell+n=9$

(C) $\ell+2m=10$

(D) $m+n=4$

40. All the 7-digit Numbers containing each of the digits 1, 2, 3, 4, 5, 6, 7 exactly once, and not divisible by 5, are arranged in increasing order. Then which of the following is/are **CORRECT**?

(A) 1993rd number in the list is 4312567

(B) 1996th number in the list is 4312756

(C) 2000th number in the list is 4315672

(D) 1999th number in the list is 43152763

41. Let $f(x)=\begin{cases}\lim\limits_{n\to\infty}\left(\dfrac{px}{n}\displaystyle\sum_{r=1}^{n}\dfrac{\left[r^2-e^{-x}+r-1\right]}{r(r+1)}\right)+\lambda, & x>0\\ q, & x=0\\ \lim\limits_{n\to\infty}\displaystyle\sum_{r=1}^{n}\dfrac{\left\{r^2+r+e^{x}-1\right\}}{r(r+1)}, & x<0\end{cases}$

Is differentiable is R:

$\left([.]\text{ is G.I.F. and }\{.\}\text{ is F.P. of x}\right)$ Then:

(A) $p=1$

(B) $q=1$

(C) $p+q+\lambda=3$

(D) if g is inverse of F then $g'(1/2)=2$

42. Consider $f(x)=4x^4-24x^3+31x^2+6x-8$ be a polynomial function and $\alpha,\beta,\gamma,\sigma$ are the roots. $(\alpha<\beta<\gamma<\sigma)$. So:

(A) $\sigma^p+\frac{1}{\sigma^\alpha}+\sigma^r+r^\sigma=36$

(B) $\sigma^p+\frac{1}{\sigma^\alpha}+\sigma^r+r^\sigma=18$

(C) $\displaystyle\int_{2\alpha}^{2\beta}\frac{x^{\sigma+1}-5x^{r+1}+2\beta|x|+1}{x^2+4\beta|x|+1}dx=2\ln 2$

(D) $\int_{2\alpha}^{2\beta} \frac{x^{\sigma+1}-5x^{r+1}+2\beta|x|+1}{x^2+4\beta|x|+1}dx = \ln 2$

43. Each of 2010 boxes in a line contains one red marble and for $1 \le k \le 2010,$ the box is the k^{th} position also contain k white marbles. A child begins at the fist box and successively drawn a single marble at random from each box in order. The stops when be fist draws a red marble. Let $p(n)$ be the probability that he stops after drawing exactly n marbles. The possible value(s) of n for which $p(n) < \frac{1}{2010}$ is:

(A) 44
(B) 45
(C) 46
(D) 47

44. Consider the equation $z^2-(3+i)z+(m+2i)=0\,(m \in R)$. If the equation has exactly one real and one-non-real complex root, then which of the following hold(s) good:

(A) Modulus of the non-real complex root is 2
(B) The value of m is 3
(C) Additive inverse of non-real root is $(-1-i)$
(D) Product of real root and imaginary part of non-real complex root is 2

Section – A (Maximum Marks: 12)

*This section contains **TWO (02) paragraphs.** Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

Paragraph for Question Nos. 45 and 46

Let t be a real number satisfying $2t^3-9t^2+30-\lambda=0$ where $t = x+\frac{1}{x}$ and $\lambda \in R$ then

45. If the cubic equation has three real and distinct solution for x then λ

(A) Is greater than 9
(B) Is greater than 11
(C) Is less than 8
(D) Is not equal to 10

46. If the cubic equation has exactly two real and distinct roots of x then exhaustive set of values of λ is

(A) $\lambda \in (-\infty,3) \cup (30,\infty)$
(B) $\lambda \in (-\infty,-22) \cup (10,\infty) \cup \{3\}$
(C) $\lambda \in \{3,30\}$
(D) None of these

Paragraph for Question Nos. 47 and 48

Four points A, B, C and D lie in that order on the parabola $y = ax^2 + bx + c$ and the coordinates of A, B and D are known $A(-2,3); B(-1,1); D(2,7)$.

47. The value of a is:
   (A) 1
   (B) 2
   (C) 3
   (D) 4

48. The value of c is:
   (A) 3
   (B) 2
   (C) 0
   (D) None of these

Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.***

49. Let $p(x)$ be a polynomial of degree 6 with leading coefficient unity and $p(-x) = p(x)\, \forall x \in R$. Also $\left(p(1)+3\right)^2 + p^2(2) + \left(p(3)-5\right)^2 = 0$ then $\sqrt{-4-p(0)}$ is____

50. Let $S = \left\{(x, y); x^2 + y^2 - 6x - 8y + 21 \le 0\right\}$ then

$$\max\left\{\frac{12x}{7} - \frac{5y}{7}; (x, y) \in S\right\} + \min\left\{\frac{1}{2}\left(x^2 + y^2 + 1\right) + (x - y); (x - y) \in S\right\}$$
$$-\min\left\{\frac{\sqrt{3}y + |x-3|}{|x-3|}; (x, y) \in S\right\}$$

51. Let $f(x) = \cos 2x.\cos 4x.\cos 6x.\cos 8x.\cos 10x$ then $\lim\limits_{x \to 0} \frac{1-\left(f(x)\right)^3}{55\sin^2 x}$ equals

Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 52 and 53

Question Stem

Mr. A randomly picks 3 distinct numbers from the set $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ and arranges them in descending order to form a three digit number. Mr. B randomly picks 3 distinct numbers from the set $\{1, 2, 3, 4, 5, 6, 7, 8\}$ and also arranges them in descending order to form a 3 digit number.

52. The probability that Mr. A's 3 digit number is always greater than Mr. B's 3 digit number is:

53. The probability that A and B has the same 3 digit number is:

Question Stem for Question Nos. 54 and 55

Question Stem

If $f:(0, \infty) \to (0, \infty)$ satisfy $f(xf(y)) = x^2 y^n$, $(n \in R)$, then

54. Value of $f(1)$ will be____

55. Number of solutions of $2f(x) = e^x$ is___

Question Stem for Question Nos. 56 and 57

Question Stem

Suppose equation is $f(x) - g(x) = 0$ or $f(x) = g(x) = y$ say, then draw the graphs of $y = f(x)$ and $y = g(x)$. If graphs of $y = f(x)$ and $y = g(x)$ cut at one, two, three, …. , no points, then number of solutions are one, two, three, ….., zero respectively.

56. The number of solutions of $\sin x = \frac{|x|}{10}$ is___

57. Total number of solutions of the equation $3x + 2\tan x = \frac{5\pi}{2}$ in $x \in [0, 2\pi]$ is equal to___